Data Makes AI Project Brief

Parent item	Unit [Shared Core 1]: Data Makes AI
Type	Design Docs Internal
Status	Done

A. Project Overview

Project Title: Something something Game Brain

Unit: SC1: Data Makes AI

Final Product Type: A trained interactive classification model (a decision engine for a game) and a visual strategy guide or model card evaluating/explaining the AI's behavior

Student Role or Framing Device: Students act as AI Developers and Auditors, creating a decision engine for a game and documenting its behavior for players and stakeholders.

When Project Begins: TBD — there is a strong case to be made that this project could begin shortly after unit launch to best accommodate all the steps of the process. It might feel like too much condensed to the end of the unit. There lots of learning moments within the structure of the project that can shrink or contract (flex points) depending on which route we want to go (traditional end-of-unit, or hybrid PBL).

B. Purpose and Skill Alignment

Why this project?

What makes this compelling, authentic, or meaningful to students?

- There are robust initial datasets that are created from a variety of different game genres in order to appeal to diverse student interests

- This project shifts students from evaluating AI models purely as observers to understanding it as active builders
- Students discover that every choice they make about what data to include, exclude, or label has real consequences on what the model learns
- Creating an original game model is more motivating to middle school students than simply fixing a broken model
- By testing the model through the lens of different stakeholder personas (a five-year-old vs. an adult), students authentically experience how bias in training data can lead to harmful or unexpected outputs

Primary Concepts Practiced

- **Data Curation & Labeling**: Making deliberate choices about what to include or exclude
- **Supervised Learning Patterns**: Explaining how models generate predictions based on training data
- **Bias Identification**: Investigating how data quality, quantity, and composition affects model accuracy and fairness
- **Iterative Refinement**: Testing, identifying failure points, and retraining models to improve accuracy

What success looks like

By the end of the project, students can:

- Build a classification model using a graphical, interactive playground
- Add and label new features to a dataset to customize model behavior
- Test their model's outputs and adjust inputs to investigate successes and failures
- Identify and explain how AI might amplify societal biases through their training data choices
- Synthesize their findings into a visual Strategy Guide (Model Card) that explains the AI's "brain" to others

C. Early Ideation and Exploration

Initial Ideas

Top 2-3 Project Ideas Considered

1. **Petting Zoo AI / Treasure Finder**: Creating simple classifiers to determine safety or value based on single patterns
2. **Club Recruiter / What Makes a Hit?**: Curating data about student interests or music trends to build a recommendation model that identifies what might be popular with peers
3. R**everse Engineering / Creating a Problem Solver**: Investigating targeted ads or building a tool to solve a community problem, such as sorting trash vs. recycling

Project Criteria Used for Evaluation

- Engagement: Does the theme (gaming/classification) appeal to middle schoolers?
- Alignment to unit objectives with the ability to act as a summative assessment
- Technical Feasibility: Can students build the model in a 3-week window?
- Authenticity: Does the project reflect how real-world data choices lead to model consequences?
- Feasibility of providing accessible, constrained datasets that still allow for high autonomy

Why This Project Was Selected

This project was selected because it:

- Provides a highly engaging, concrete context (gaming) where testing edge cases and evaluating outcomes feels authentic and motivating to middle schoolers

- Clearly maps data curation choices (what to include, exclude, or label) to direct consequences in the model's behavior
- Allows for natural integration of stakeholder testing to uncover bias without being overly abstract

Tradeoff: We intentionally chose a semi-structured "library of assets" approach over a completely open-ended research project to ensure students focus on the mechanics of supervised learning rather than getting bogged down in the logistics of data collection.

D. Design Variants Explored

Alternate Version #1

AI Auditor / Fix the Game (Stretch the Theme)

Students act as an AI Auditor for a major game studio. The studio's "Hero Selector" is reporting high rates of biased outcomes. Students must reverse-engineer the existing model by tracing predictions, find where the bias is hiding, and re-train the model with a fairer dataset

Why not selected: The narrative felt too restrictive and capitalistic (fixing a game for a company so it sells better). Middle schoolers are generally more motivated by the autonomy of creating their own original game model from scratch rather than just fixing someone else's broken one

Alternate Version #2

Strategy Guide (Tweak the Medium)

Instead of solely building a model, students create a visual "Strategy Guide" (like Nintendo Power) for players to understand how the game's AI makes its choices. The final product is a visual representation mapping out decision paths for different types of players, comparing human-made rules to AI-generated rules.

Why not selected: As a standalone project, it felt too similar to a standard report or model card. It risked removing the interactive, iterative building and testing components that make the unit hands-on.

Incorporated Elements from Variants

What did you keep or remix from these versions?

- **High Autonomy Asset Library**: We kept the core idea of giving students a rich library of data assets to build their models from scratch, rather than giving them a pre-built model
- **The Strategy Guide as a Synthesis Tool**: We integrated the Strategy Guide concept as the project's final synthesis/reflection component (serving as a visual model card) rather than making it the entire project
- **Stakeholder Testing**: We retained the focus on testing for bias from the AI Auditor variant, but remixed it so students test their own created models by assuming different stakeholder personas

E. Risks, Gaps, or Open Questions

Potential Risks or Weak Spots

- **Collaboration vs. Ownership**: There is a risk that group work could result in one student doing the majority of the work. To mitigate this, students must maintain individual ownership of their models while using group time to competitively test each other's models
- **Technical Constraints**: True pair programming is not supported in the current IDE/tools being used, which limits real-time collaborative building
- **Tedium in Data Labeling**: If the data labeling process is too manual or time-consuming, it will become tedious. The tool needs to provide instant, repetitive feedback to keep students engaged

Devil’s Advocate Highlights

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Open Questions for Reviewers

- How can we visualize the *features* within data and the model's *decision trees* clearly?
- Should the AI tool natively assist students in suggesting and auto-filling features to reduce the burden of manual data entry?

F. Project Components

What Students Produce

How Project Work Is Distributed Across the Unit

1. Students choose a dataset of game assets, define a clear classification goal, and examine the labeled categories the model will learn from.
2. Students train a first-pass classification model using the dataset's existing features and make predictions about what patterns the model will learn.
3. Students evaluate the model by testing it on new examples, identifying where it succeeds or makes mistakes, and analyzing its performance from different stakeholder perspectives.
4. Students modify the model by adding or refining features, retrain it, and analyze how those changes impact its predictions and overall performance.
5. Students create a visual "Strategy Guide" (model card) that explains what the model predicts, the features it uses, the patterns it learned, and its limitations or biases.

Final Deliverable:

- **A Trained Classification Model:** A model that predicts categories for game assets by learning patterns from labeled data and applying them to new, unseen examples.
- **A "Strategy Guide" / Model Card:** A visual explanation of how the model works, including the features it uses, the patterns it learned, examples of its predictions, and its limitations or potential biases.
- **Stakeholder Feedback Report:** An analysis of how the model performs for different users, identifying who it works well for, who may be excluded, and

how it could be improved.

How It’s Assessed

- Project Rubric

Category	Exceeds	Meets	Approaching	Beginning
Defining the Problem & Data	I clearly defined what my model should predict. I chose useful features and explained why they matter. My labels are clear, consistent, and make sense.	I defined what my model predicts. I used appropriate features and labels.	My prediction goal, features, or labels are somewhat unclear.	My model setup is unclear or incomplete.
How the Model Works	I clearly explain how my model makes predictions step-by-step. I explain how the model learned from data (not just rules I made).	I explain how my model makes predictions using rules. I understand that the model learned from data.	My explanation is incomplete or confusing.	I cannot explain how my model makes predictions.
Testing & Improving	I tested my model with different examples (including tricky ones). I found patterns in mistakes and improved my model. I clearly explain how my changes made it better.	I tested my model and made at least one improvement. I explain how my changes affected results.	I did limited testing or made changes without clear reasoning.	I did not test or improve my model.
Fairness & Impact	I clearly explain who my model works well for and who it doesn’t. I connect problems to data choices and suggest ways to improve fairness.	I identify who the model works for and who it might not. I recognize that data can cause bias.	I mention fairness but don’t explain it clearly.	I do not consider fairness or impact.
Explaining Your Model (Strategy Guide)	My explanation is clear, organized, and easy to understand. I include features, predictions, mistakes, and limitations.	I explain how my model works, including features and predictions. I include some limitations.	My explanation is missing key parts or is hard to follow.	My explanation is unclear or incomplete.

Supports Students will Get

Links and Artifacts

- Possible Learning Trajectories

Concept	Stage 0 (Missing)	Stage 1 (Beginning)	Stage 2 (Meeting)	Stage 3 (Exceeding)
Machine Learning Logic *Students learn to curate datasets, train classification models, test their outputs, and evaluate their limitations using a structured, iterative process* Stage 0 → Stage 1 Stage 1 → Stage 2 Stage 2 → Stage 3	• Adds or removes data randomly without a clear goal • Treats the model like magic, assuming it just knows the answers, rather than learning from the provided examples • Stops working once the model gives a single result, without checking if it actually works well	• Follows instructions to train a model but relies on trial-and-error to fix it when it gets things wrong • Notices when the model makes a mistake, but struggles to explain how their specific data caused that mistake • Relies heavily on examples or teacher help to figure out what data to add next	• Chooses data deliberately to teach the model a specific concept or rule • Explains that the model makes decisions by finding patterns in the examples it was given • Tests the model on purpose to see where it gets confused and where it gets things right • Improves the model by going back and changing the examples (adding more, removing confusing ones, or fixing labels)	• Predicts where the model will fail before testing it, based on what they know is missing from their data • Invents new, creative categories or features (labels) to help the model understand tricky edge cases that a standard dataset misses • Clearly explains the limits of their model
Algorithmic Bias *Students investigate how the quality, quantity, and composition of training data affects a model's fairness, and identify ways that AI systems can amplify societal biases* Stage 0 → Stage 1 Stage 1 → Stage 2 Stage 2 → Stage 3	• Cares only about whether the model is accurate overall, ignoring how it might treat different types of users • Assumes that because a computer made the decision, the outcome is automatically fair and unbiased	• Spots unfairness or bias only when a teacher points it out or when looking at a really obvious example • Knows that data causes bias, but cannot explain how their own choices of data might cause bias	• Tests a model on purpose by pretending to be different types of users to see who it might leave out • Connects a biased or unfair result directly back to a choice they made • Explains how their specific model could cause frustration or harm to certain groups of users	• Identifies potential bias in the provided dataset before training the model, and takes steps to fix it early • Articulates the trade-offs of their choices, understanding that trying to make the model perfect for one group might accidentally make it worse for another • Explains how the absence of data (who or what is missing) is just as powerful as the data that is included

CPM Notes

⚖️ **Tradeoffs We're Accepting**
List any limitations or design compromises the team knowingly accepted when selecting or scoping this project.

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▼ CPM Review

Pre-Check: Has the team completed all required sections?

- ☐ Project title + product type
- ☐ Student role or framing device
- ☐ When project begins
- ☐ Primary concepts from Skills and Concepts Framework
- ☐ Initial idea generation + team brainstorming
- ☐ Project selection rationale + criteria
- ☐ Design variants explored
- ☐ Risks + Devil's Advocate section
- ☐ Final product + assessment summary
- ☐ Links to relevant scaffolds or rubric (if applicable)

Core Review Questions

A. Alignment and Purpose

Does the project clearly support the Skills and Concepts Framework concepts?

Is the purpose of the project meaningful and student-relevant?

Does this project assess a student's ability to perform at the target cognitive demand levels from the Skills and Concepts Framework?

Would this project surface real evidence of student thinking on the targeted skills?

B. Feasibility and Fit

Is the scope realistic for the planned number of lessons?

Can it be executed in a variety of classroom settings (tech, time, ability)?

C. Design Rigor

Did the team explore multiple viable ideas before selecting this one?

Do the design variants show thoughtful iteration?

Are tradeoffs acknowledged and risks named?

D. Assessment Readiness

Is it clear how this project will be evaluated (e.g., rubric, checklist)?

Will the evaluation allow differentiation of student skill levels?

Feedback

What’s strong

What needs clarification or improvement

Suggestions

Final Signoff

- [ ] Approved for Lesson Progression Planning

Notes:

- [ ] Needs Revisions

Action Steps: